

The caveat at the end of the above theorem is necessary — there are certainly cases in which the derivative does not exist and so we do need to be careful.

Example 2.3.3

Find the tangent line to the curve $y = \sqrt{x}$ at $x = 4$.

Rather than redoing everything from scratch, we can, and for efficiency, should, use Theorem 2.3.2. To write this up properly, we must ensure that we tell the reader what we are doing. So something like the following:

- By Theorem 2.3.2, the tangent line to the curve $y = f(x)$ at $x = a$ is given by

$$y = f(a) + f'(a)(x - a)$$

provided $f'(a)$ exists.

- In Example 2.2.9, we found that, for any $a > 0$, the derivative of $\sqrt{x}$ at $x = a$ is

$$f'(a) = \frac{1}{2\sqrt{a}}$$

- In the current example, $a = 4$ and we have

$$f(a) = f(4) = \sqrt{x}|_{x=4} = \sqrt{4} = 2 \quad \text{and} \quad f'(a) = f'(4) = \frac{1}{2\sqrt{a}}|_{a=4} = \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

- So the equation of the tangent line to $y = \sqrt{x}$ at $x = 4$ is

$$y = 2 + \frac{1}{4}(x - 4) \quad \text{or} \quad y = \frac{x}{4} + 1$$

We don't have to write it up using dot-points as above; we have used them here to help delineate each step in the process of computing the tangent line.

Example 2.3.3

2.4 ▲ Arithmetic of Derivatives - a Differentiation Toolbox

So far, we have evaluated derivatives only by applying Definition 2.2.1 to the function at hand and then computing the required limits directly. It is quite obvious that as the function being differentiated becomes even a little complicated, this procedure quickly becomes extremely unwieldy. It is many orders of magnitude more efficient to have access to

- a list of derivatives of some simple functions and
- a collection of rules for breaking down complicated derivative computations into sequences of simple derivative computations.

This is precisely what we did to compute limits. We started with limits of simple functions and then used “arithmetic of limits” to compute limits of complicated functions.

We have already started building our list of derivatives of simple functions. We have shown, in Examples [2.2.2](#), [2.2.3](#), [2.2.5](#) and [2.2.9](#), that

$$\frac{d}{dx}1 = 0 \qquad \frac{d}{dx}x = 1 \qquad \frac{d}{dx}x^2 = 2x \qquad \frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$$

We’ll expand this list later.

We now start building a collection of tools that help reduce the problem of computing the derivative of a complicated function to that of computing the derivatives of a number of simple functions. In this section we give three derivative “rules” as three separate theorems. We’ll give the proofs of these theorems in the next section and examples of how they are used in the following section.

As was the case for limits, derivatives interact very cleanly with addition, subtraction and multiplication by a constant. The following result actually follows very directly from the first three points of Theorem [1.4.2](#).

Lemma 2.4.1 (Derivative of sum and difference).

Let $f(x), g(x)$ be differentiable functions and let $c \in \mathbb{R}$ be a constant. Then

$$\begin{aligned}\frac{d}{dx}\{f(x) + g(x)\} &= f'(x) + g'(x) \\ \frac{d}{dx}\{f(x) - g(x)\} &= f'(x) - g'(x) \\ \frac{d}{dx}\{cf(x)\} &= cf'(x)\end{aligned}$$

That is, the derivative of the sum is the sum of the derivatives, and so forth.

Following this we can combine the three statements in this lemma into a single rule which captures the “linearity of differentiation”.

Theorem 2.4.2 (Linearity of differentiation).

Again, let $f(x), g(x)$ be differentiable functions, let $\alpha, \beta \in \mathbb{R}$ be constants and define the “linear combination”

$$S(x) = \alpha f(x) + \beta g(x).$$

Then the derivative of $S(x)$ at $x = a$ exists and is

$$\frac{dS}{dx} = S'(x) = \alpha f'(x) + \beta g'(x).$$

Note that we can recover the three rules in the previous lemma by setting $\alpha = \beta = 1$ or $\alpha = 1, \beta = -1$ or $\alpha = c, \beta = 0$.

Unfortunately, the derivative does not act quite as simply on products or quotients. The rules for computing derivatives of products and quotients get their own names and theorems:

Theorem 2.4.3 (The product rule).

Let $f(x), g(x)$ be differentiable functions, then the derivative of the product $f(x)g(x)$ exists and is given by

$$\frac{d}{dx}\{f(x)g(x)\} = f'(x)g(x) + f(x)g'(x).$$

Before we proceed to the derivative of the ratio of two functions, it is worth noting a special case of the product rule when $g(x) = f(x)$. In fact, since this is a useful special case, let us call it a corollary¹⁴.

Corollary 2.4.4 (Derivative of a square).

Let $f(x)$ be a differentiable function, then the derivative of its square is:

$$\frac{d}{dx}\{f(x)^2\} = 2f(x)f'(x)$$

With a little work this can be generalised to other powers — but that is best done once we understand how to compute the derivative of the composition of two functions. That requires the chain rule (see Theorem 2.9.2 below). But before we get to that, we need to see how to take the derivative of a quotient of two functions.

Theorem 2.4.5 (The quotient rule).

Let $f(x), g(x)$ be differentiable functions. Then the derivative of their quotient is

$$\frac{d}{dx}\left\{\frac{f(x)}{g(x)}\right\} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

This derivative exists except at points where $g(x) = 0$.

There is a useful special case of this theorem which we obtain by setting $f(x) = 1$. In that case, the quotient rule tells us how to compute the derivative of the reciprocal of a function.

14 Recall that a corollary is an important result that follows from one or more theorems — typically without too much extra work — as is the case here.

Corollary 2.4.6 (Derivative of a reciprocal).

Let $g(x)$ be a differentiable function. Then the derivative of the reciprocal of g is given by

$$\frac{d}{dx} \left\{ \frac{1}{g(x)} \right\} = -\frac{g'(x)}{g(x)^2}$$

and exists except at those points where $g(x) = 0$.

So we have covered, sums, differences, products and quotients. This allows us to compute derivatives of many different functions — including polynomials and rational functions. However we are still missing trigonometric functions (for example), and a rule for computing derivatives of compositions. These will follow in the near future, but there are a couple of things to do before that — understand where the above theorems come from, and practice using them.

2.5 ▲ Proofs of the Arithmetic of Derivatives

The theorems of the previous section are not too difficult to prove from the definition of the derivative (which we know) and the arithmetic of limits (which we also know). In this section we show how to construct these rules.

Throughout this section we will use our two functions $f(x)$ and $g(x)$. Since the theorems we are going to prove all express derivatives of linear combinations, products and quotients in terms of f, g and their derivatives, it is helpful to recall the definitions of the derivatives of f and g :

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}.$$

Our proofs, roughly speaking, involve doing algebraic manipulations to uncover the expressions that look like the above.

► Proof of the Linearity of Differentiation (Theorem 2.4.2)

Recall that in Theorem 2.4.2 we defined $S(x) = \alpha f(x) + \beta g(x)$, where $\alpha, \beta \in \mathbb{R}$ are constants. We wish to compute $S'(x)$, so we start with the definition:

$$S'(x) = \lim_{h \rightarrow 0} \frac{S(x+h) - S(x)}{h}$$

Let us concentrate on the numerator of the expression inside the limit and then come back to the full limit in a moment. Substitute in the definition of $S(x)$:

$$\begin{aligned} S(x+h) - S(x) &= [\alpha f(x+h) + \beta g(x+h)] - [\alpha f(x) + \beta g(x)] && \text{collect terms} \\ &= \alpha [f(x+h) - f(x)] + \beta [g(x+h) - g(x)] \end{aligned}$$

Now it is easy to see the structures we need — namely, we almost have the expressions for the derivatives $f'(x)$ and $g'(x)$. Indeed, all we need to do is divide by h and take the limit. So let's finish things off.

$$\begin{aligned}
 S'(x) &= \lim_{h \rightarrow 0} \frac{S(x+h) - S(x)}{h} && \text{from above} \\
 &= \lim_{h \rightarrow 0} \frac{\alpha[f(x+h) - f(x)] + \beta[g(x+h) - g(x)]}{h} \\
 &= \lim_{h \rightarrow 0} \left[\alpha \frac{f(x+h) - f(x)}{h} + \beta \frac{g(x+h) - g(x)}{h} \right] && \text{limit laws} \\
 &= \alpha \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \beta \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\
 &= \alpha f'(x) + \beta g'(x)
 \end{aligned}$$

as required.

► Proof of the Product Rule (Theorem 2.4.3)

After the warm-up above, we will just jump straight in. Let $P(x) = f(x)g(x)$, the product of our two functions. The derivative of the product is given by

$$P'(x) = \lim_{h \rightarrow 0} \frac{P(x+h) - P(x)}{h}$$

Again we will focus on the numerator inside the limit and massage it into the form we need. To simplify these manipulations, define

$$F(h) = \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad G(h) = \frac{g(x+h) - g(x)}{h}.$$

Then we can write

$$f(x+h) = f(x) + hF(h) \quad \text{and} \quad g(x+h) = g(x) + hG(h).$$

We can also write

$$f'(x) = \lim_{h \rightarrow 0} F(h) \quad \text{and} \quad g'(x) = \lim_{h \rightarrow 0} G(h).$$

So back to that numerator:

$$\begin{aligned}
 P(x+h) - P(x) &= f(x+h) \cdot g(x+h) - f(x) \cdot g(x) && \text{substitute} \\
 &= [f(x) + hF(h)] [g(x) + hG(h)] - f(x) \cdot g(x) && \text{expand} \\
 &= f(x)g(x) + f(x) \cdot hG(h) + hF(h) \cdot g(x) + h^2F(h) \cdot G(h) - f(x) \cdot g(x) \\
 &= f(x) \cdot hG(h) + hF(h) \cdot g(x) + h^2F(h) \cdot G(h).
 \end{aligned}$$

Armed with this we return to the definition of the derivative:

$$\begin{aligned}
 P'(x) &= \lim_{h \rightarrow 0} \frac{P(x+h) - P(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x) \cdot hG(h) + hF(h) \cdot g(x) + h^2F(h) \cdot G(h)}{h} \\
 &= \left(\lim_{h \rightarrow 0} \frac{f(x) \cdot hG(h)}{h} \right) + \left(\lim_{h \rightarrow 0} \frac{hF(h) \cdot g(x)}{h} \right) + \left(\lim_{h \rightarrow 0} \frac{h^2F(h) \cdot G(h)}{h} \right) \\
 &= \left(\lim_{h \rightarrow 0} f(x) \cdot G(h) \right) + \left(\lim_{h \rightarrow 0} F(h) \cdot g(x) \right) + \left(\lim_{h \rightarrow 0} hF(h) \cdot G(h) \right)
 \end{aligned}$$

Now since $f(x)$ and $g(x)$ do not change as we send h to zero, we can pull them outside. We can also write the third term as the product of 3 limits:

$$\begin{aligned}
 &= \left(f(x) \lim_{h \rightarrow 0} G(h) \right) + \left(g(x) \lim_{h \rightarrow 0} F(h) \right) + \left(\lim_{h \rightarrow 0} h \right) \cdot \left(\lim_{h \rightarrow 0} F(h) \right) \cdot \left(\lim_{h \rightarrow 0} G(h) \right) \\
 &= f(x) \cdot g'(x) + g(x) \cdot f'(x) + 0 \cdot f'(x) \cdot g'(x) \\
 &= f(x) \cdot g'(x) + g(x) \cdot f'(x).
 \end{aligned}$$

And so we recover the product rule.

► (optional) — Proof of the Quotient Rule (Theorem 2.4.5)

We now give the proof of the quotient rule in two steps¹⁵. We assume throughout that $g(x) \neq 0$ and that $f(x)$ and $g(x)$ are differentiable, meaning that the limits defining $f'(x)$, $g'(x)$ exist.

- In the first step, we prove the quotient rule under the assumption that $f(x)/g(x)$ is differentiable.
- In the second step, we prove that $1/g(x)$ is differentiable. Once we know that $1/g(x)$ is differentiable, the product rule implies that $f(x)/g(x)$ is differentiable.

Step 1: the proof of the quotient rule assuming that $\frac{f(x)}{g(x)}$ is differentiable. Write $Q(x) = \frac{f(x)}{g(x)}$. Then $f(x) = g(x) Q(x)$ so that $f'(x) = g'(x) Q(x) + g(x) Q'(x)$, by the product rule, and

$$\begin{aligned}
 Q'(x) &= \frac{f'(x) - g'(x) Q(x)}{g(x)} = \frac{f'(x) - g'(x) \frac{f(x)}{g(x)}}{g(x)} \\
 &= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}
 \end{aligned}$$

¹⁵ We thank Serban Raianu for suggesting this approach.

Step 2: the proof that $1/g(x)$ is differentiable. By definition

$$\begin{aligned}\frac{d}{dx} \frac{1}{g(x)} &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{1}{g(x+h)} - \frac{1}{g(x)} \right] = \lim_{h \rightarrow 0} \frac{g(x) - g(x+h)}{h g(x) g(x+h)} \\ &= - \lim_{h \rightarrow 0} \frac{1}{g(x)} \frac{1}{g(x+h)} \frac{g(x+h) - g(x)}{h} \\ &= - \frac{1}{g(x)} \lim_{h \rightarrow 0} \frac{1}{g(x+h)} \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= - \frac{1}{g(x)^2} g'(x)\end{aligned}$$

2.6 ▲ Using the Arithmetic of Derivatives – Examples

In this section we illustrate the computation of derivatives using the arithmetic of derivatives — Theorems [2.4.2](#), [2.4.3](#) and [2.4.5](#). To make it clear which rules we are using during the examples we will note which theorem we are using:

- LIN to stand for “linearity” $\frac{d}{dx} \{ \alpha f(x) + \beta g(x) \} = \alpha f'(x) + \beta g'(x)$ Theorem [2.4.2](#)
- PR to stand for “product rule” $\frac{d}{dx} \{ f(x) g(x) \} = f'(x) g(x) + f(x) g'(x)$ Theorem [2.4.3](#)
- QR to stand for “quotient rule” $\frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{f'(x) g(x) - f(x) g'(x)}{g(x)^2}$ Theorem [2.4.5](#)

We’ll start with a really easy example.

Example 2.6.1

$$\begin{aligned}\frac{d}{dx} \{4x + 7\} &= 4 \cdot \frac{d}{dx} \{x\} + 7 \cdot \frac{d}{dx} \{1\} && \text{LIN} \\ &= 4 \cdot 1 + 7 \cdot 0 = 4\end{aligned}$$

where we have used LIN with $f(x) = x$, $g(x) = 1$, $\alpha = 4$, $\beta = 7$.

Example 2.6.1

Example 2.6.2

Continuing on from the previous example, we can use the product rule and the previous result to compute

$$\begin{aligned}\frac{d}{dx} \{x(4x + 7)\} &= x \cdot \frac{d}{dx} \{4x + 7\} + (4x + 7) \frac{d}{dx} \{x\} && \text{PR} \\ &= x \cdot 4 + (4x + 7) \cdot 1 \\ &= 8x + 7\end{aligned}$$

where we have used the product rule PR with $f(x) = x$ and $g(x) = 4x + 7$.

Example 2.6.2

Example 2.6.3

In the same vein as the previous example, we can use the quotient rule to compute

$$\begin{aligned}\frac{d}{dx} \left\{ \frac{x}{4x+7} \right\} &= \frac{(4x+7) \cdot \frac{d}{dx}\{x\} - x \cdot \frac{d}{dx}\{4x+7\}}{(4x+7)^2} && \text{QR} \\ &= \frac{(4x+7) \cdot 1 - x \cdot 4}{(4x+7)^2} \\ &= \frac{7}{(4x+7)^2}\end{aligned}$$

where we have used the quotient rule QR with $f(x) = x$ and $g(x) = 4x + 7$.

Example 2.6.3

Now for a messier example.

Example 2.6.4

Differentiate

$$f(x) = \frac{x}{2x + \frac{1}{3x+1}}$$

This problem looks nasty. But it isn't so hard if we just build it up a bit at a time.

- First, $f(x)$ is the ratio of

$$f_1(x) = x \quad \text{and} \quad f_2(x) = 2x + \frac{1}{3x+1}$$

If we can find the derivatives of $f_1(x)$ and $f_2(x)$, we will be able to get the derivative of $f(x)$ just by applying the quotient rule. The derivative, $f'_1(x) = 1$, of $f_1(x)$ is easy, so let's work on $f_2(x)$.

- The function $f_2(x)$ is the linear combination

$$f_2(x) = 2f_3(x) + f_4(x) \quad \text{with} \quad f_3(x) = x \quad \text{and} \quad f_4(x) = \frac{1}{3x+1}$$

If we can find the derivatives of $f_3(x)$ and $f_4(x)$, we will be able to get the derivative of $f_2(x)$ just by applying linearity (Theorem [2.4.2](#)). The derivative, $f'_3(x) = 1$, of $f_3(x)$ is easy. So let's work on $f_4(x)$.

- The function $f_4(x)$ is the ratio

$$f_4(x) = \frac{1}{f_5(x)} \quad \text{with} \quad f_5(x) = 3x + 1$$

If we can find the derivative of $f_5(x)$, we will be able to get the derivative of $f_4(x)$ just by applying the special case the quotient rule (Corollary [2.4.6](#)). The derivative of $f_5(x)$ is easy.

- So we have completed breaking down $f(x)$ into easy pieces. It is now just a matter of reversing the break down steps, putting everything back together, starting with the easy pieces and working up to $f(x)$. Here goes.

$$f_5(x) = 3x + 1 \quad \text{so } \frac{d}{dx}f_5(x) = 3\frac{d}{dx}x + \frac{d}{dx}1 = 3 \cdot 1 + 0 = 3 \quad \text{LIN}$$

$$f_4(x) = \frac{1}{f_5(x)} \quad \text{so } \frac{d}{dx}f_4(x) = -\frac{f_5'(x)}{f_5(x)^2} = -\frac{3}{(3x+1)^2} \quad \text{QR}$$

$$f_2(x) = 2f_3(x) + f_4(x) \quad \text{so } \frac{d}{dx}f_2(x) = 2f_3'(x) + f_4'(x) = 2 - \frac{3}{(3x+1)^2} \quad \text{LIN}$$

$$\begin{aligned} f(x) = \frac{f_1(x)}{f_2(x)} \quad \text{so } \frac{d}{dx}f(x) &= \frac{f_1'(x)f_2(x) - f_1(x)f_2'(x)}{f_2(x)^2} \quad \text{QR} \\ &= \frac{1\left[2x + \frac{1}{3x+1}\right] - x\left[2 - \frac{3}{(3x+1)^2}\right]}{\left[2x + \frac{1}{3x+1}\right]^2} \end{aligned}$$

Oof!

- We now have an answer. But we really should clean it up, not only to make it easier to read, but also because invariably such computations are just small steps inside much larger computations. Any future computations involving this expression will be a lot easier and less error prone if we clean it up now. Cancelling the $2x$ and the $-2x$ in

$$\begin{aligned} 1\left[2x + \frac{1}{3x+1}\right] - x\left[2 - \frac{3}{(3x+1)^2}\right] &= 2x + \frac{1}{3x+1} - 2x + \frac{3x}{(3x+1)^2} \\ &= \frac{1}{3x+1} + \frac{3x}{(3x+1)^2} \end{aligned}$$

and multiplying both the numerator and denominator by $(3x+1)^2$ gives

$$\begin{aligned} f'(x) &= \frac{\frac{1}{3x+1} + \frac{3x}{(3x+1)^2}}{\left[2x + \frac{1}{3x+1}\right]^2} \frac{(3x+1)^2}{(3x+1)^2} \\ &= \frac{(3x+1) + 3x}{[2x(3x+1) + 1]^2} \\ &= \frac{6x+1}{[6x^2 + 2x + 1]^2}. \end{aligned}$$

Example 2.6.4

While the linearity theorem (Theorem [2.4.2](#)) is stated for a linear combination of two functions, it is not difficult to extend it to linear combinations of three or more functions as the following example shows.

Example 2.6.5

We'll start by generalising linearity to three functions.

$$\begin{aligned}
 \frac{d}{dx}\{aF(x) + bG(x) + cH(x)\} &= \frac{d}{dx}\{a \cdot [F(x)] + 1 \cdot [bG(x) + cH(x)]\} \\
 &= aF'(x) + \frac{d}{dx}\{bG(x) + cH(x)\} \\
 &\quad \text{by LIN with } \alpha = a, f(x) = F(x), \beta = 1, \\
 &\quad \text{and } g(x) = bG(x) + cH(x) \\
 &= aF'(x) + bG'(x) + cH'(x) \\
 &\quad \text{by LIN with } \alpha = b, f(x) = G(x), \beta = c, \\
 &\quad \text{and } g(x) = H(x)
 \end{aligned}$$

This gives us linearity for three terms, namely (just replacing upper case names by lower case names)

$$\frac{d}{dx}\{af(x) + bg(x) + ch(x)\} = af'(x) + bg'(x) + ch'(x)$$

Just by repeating the above argument many times, we may generalise to linearity for n terms, for any natural number n :

$$\frac{d}{dx}\{a_1f_1(x) + a_2f_2(x) + \cdots + a_nf_n(x)\} = a_1f'_1(x) + a_2f'_2(x) + \cdots + a_nf'_n(x)$$

Example 2.6.5

Similarly, while the product rule is stated for the product of two functions, it is not difficult to extend it to the product of three or more functions as the following example shows.

Example 2.6.6

Once again, we'll start by generalising the product rule to three factors.

$$\begin{aligned}
 \frac{d}{dx}\{F(x) G(x) H(x)\} &= F'(x) G(x) H(x) + F(x) \frac{d}{dx}\{G(x) H(x)\} \\
 &\quad \text{by PR with } f(x) = F(x) \text{ and } g(x) = G(x)H(x) \\
 &= F'(x) G(x) H(x) + F(x) \{G'(x) H(x) + G(x) H'(x)\} \\
 &\quad \text{by PR with } f(x) = G(x) \text{ and } g(x) = H(x)
 \end{aligned}$$

This gives us a product rule for three factors, namely (just replacing upper case names by lower case names)

$$\frac{d}{dx}\{f(x) g(x) h(x)\} = f'(x) g(x) h(x) + f(x) g'(x) h(x) + f(x) g(x) h'(x)$$

Observe that when we differentiate a product of three factors, the answer is a sum of three terms and in each term the derivative acts on exactly one of the original factors. Just by

repeating the above argument many times, we may generalise the product rule to give the derivative of a product of n factors, for any natural number n :

$$\begin{aligned} \frac{d}{dx}\{f_1(x) f_2(x) \cdots f_n(x)\} &= f'_1(x) f_2(x) \cdots f_n(x) \\ &\quad + f_1(x) f'_2(x) \cdots f_n(x) \\ &\quad \vdots \\ &\quad + f_1(x) f_2(x) \cdots f'_n(x) \end{aligned}$$

We can also write the above as

$$\frac{d}{dx}\{f_1(x) f_2(x) \cdots f_n(x)\} = \left[\frac{f'_1(x)}{f_1(x)} + \frac{f'_2(x)}{f_2(x)} + \cdots + \frac{f'_n(x)}{f_n(x)} \right] \cdot f_1(x) f_2(x) \cdots f_n(x)$$

When we differentiate a product of n factors, the answer is a sum of n terms and in each term the derivative acts on exactly one of the original factors. In the first term, the derivative acts on the first of the original factors. In the second term, the derivative acts on the second of the original factors. And so on.

If we make $f_1(x) = f_2(x) = \cdots = f_n(x) = f(x)$ then each of the n terms on the right hand side of the above equation is the product of $f'(x)$ and exactly $n - 1$ $f(x)$'s, and so is exactly $f(x)^{n-1} f'(x)$. So we get the following useful result

$$\frac{d}{dx} f(x)^n = n \cdot f(x)^{n-1} \cdot f'(x).$$

Example 2.6.6

This last result is quite useful, so let us write it as a lemma for future reference.

Lemma 2.6.7.

Let n be a natural number and f be a differentiable function. Then

$$\frac{d}{dx} f(x)^n = n \cdot f(x)^{n-1} \cdot f'(x).$$

This immediately gives us another useful result.

Example 2.6.8

We can now compute the derivative of x^n for any natural number n . Start with Lemma [2.6.7](#) and substitute $f(x) = x$ and $f'(x) = 1$:

$$\frac{d}{dx} x^n = n \cdot x^{n-1} \cdot 1 = n x^{n-1}$$

Example 2.6.8

Again — this is a result we will come back to quite a few times in the future, so we should make sure we can refer to it easily. However, at present this statement only holds when n is a positive integer. With a little more work we can extend this to compute x^q where q is any positive rational number and then any rational number at all (positive or negative). So let us hold off for a little longer. Instead we can make it a lemma, since it will be an ingredient in quite a few of the examples following below and in constructing the final corollary.

Lemma 2.6.9 (Derivative of x^n).

Let n be a positive integer then

$$\frac{d}{dx}x^n = nx^{n-1} \quad (2.6.1)$$

Back to more examples.

Example 2.6.10

$$\begin{aligned} \frac{d}{dx}\{2x^3 + 4x^5\} &= 2\frac{d}{dx}\{x^3\} + 4\frac{d}{dx}\{x^5\} \\ &\quad \text{by LIN with } \alpha = 2, f(x) = x^3, \beta = 4, \text{ and } g(x) = x^5 \\ &= 2\{3x^2\} + 4\{5x^4\} \\ &\quad \text{by Lemma 2.6.9, once with } n = 3, \text{ and once with } n = 5 \\ &= 6x^2 + 20x^4 \end{aligned}$$

Example 2.6.10

Example 2.6.11

In this example we'll compute $\frac{d}{dx}\{(3x + 9)(x^2 + 4x^3)\}$ in two different ways. For the first, we'll start with the product rule.

$$\begin{aligned} \frac{d}{dx}\{(3x + 9)(x^2 + 4x^3)\} &= \left\{\frac{d}{dx}(3x + 9)\right\}(x^2 + 4x^3) + (3x + 9)\frac{d}{dx}\{x^2 + 4x^3\} \\ &= \{3 \times 1 + 9 \times 0\}(x^2 + 4x^3) + (3x + 9)\{2x + 4(3x^2)\} \\ &= 3(x^2 + 4x^3) + (3x + 9)(2x + 12x^2) \\ &= 3x^2 + 12x^3 + (6x^2 + 18x + 36x^3 + 108x^2) \\ &= 18x + 117x^2 + 48x^3 \end{aligned}$$

For the second, we expand the product first and then differentiate.

$$\begin{aligned}\frac{d}{dx}\{(3x+9)(x^2+4x^3)\} &= \frac{d}{dx}\{9x^2+39x^3+12x^4\} \\ &= 9(2x) + 39(3x^2) + 12(4x^3) \\ &= 18x + 117x^2 + 48x^3\end{aligned}$$

Example 2.6.11

Example 2.6.12

$$\begin{aligned}\frac{d}{dx}\left\{\frac{4x^3-7x}{4x^2+1}\right\} &= \frac{(12x^2-7)(4x^2+1) - (4x^3-7x)(8x)}{(4x^2+1)^2} \\ &\quad \text{by QR with } f(x) = 4x^3-7x, f'(x) = 12x^2-7, \\ &\quad \text{and } g(x) = 4x^2+1, g'(x) = 8x \\ &= \frac{(48x^4-16x^2-7) - (32x^4-56x^2)}{(4x^2+1)^2} \\ &= \frac{16x^4+40x^2-7}{(4x^2+1)^2}\end{aligned}$$

Example 2.6.12

Example 2.6.13

In this example, we'll use a little trickery to find the derivative of $\sqrt[3]{x}$. The trickery consists of observing that, by the definition of the cube root,

$$x = (\sqrt[3]{x})^3.$$

Since both sides of the expression are the same, they must have the same derivatives:

$$\frac{d}{dx}\{x\} = \frac{d}{dx}(\sqrt[3]{x})^3.$$

We already know by Theorem [2.2.4](#) that

$$\frac{d}{dx}\{x\} = 1$$

and that, by Lemma [2.6.7](#) with $n = 3$ and $f(x) = \sqrt[3]{x}$,

$$\frac{d}{dx}(\sqrt[3]{x})^3 = 3(\sqrt[3]{x})^2 \cdot \frac{d}{dx}\{\sqrt[3]{x}\} = 3x^{2/3} \cdot \frac{d}{dx}\{\sqrt[3]{x}\}.$$

Since we know that $\frac{d}{dx}\{x\} = \frac{d}{dx}(\sqrt[3]{x})^3$, we must have

$$1 = 3x^{2/3} \cdot \frac{d}{dx}\{\sqrt[3]{x}\}$$

which we can rearrange to give the result we need

$$\frac{d}{dx}\{\sqrt[3]{x}\} = \frac{1}{3}x^{-2/3}$$

Example 2.6.13

Example 2.6.14

In this example, we'll use the same trickery as in Example 2.6.13 to find the derivative $x^{p/q}$ for any two natural numbers p and q . By definition of the q^{th} root,

$$x^p = (x^{p/q})^q.$$

That is, x^p and $(x^{p/q})^q$ are the same function, and so have the same derivative. So we differentiate both of them. We already know that, by Lemma 2.6.9 with $n = p$,

$$\frac{d}{dx}\{x^p\} = px^{p-1}$$

and that, by Lemma 2.6.7 with $n = q$ and $f(x) = x^{p/q}$,

$$\frac{d}{dx}\{(x^{p/q})^q\} = q(x^{p/q})^{q-1} \frac{d}{dx}\{x^{p/q}\}$$

Remember that $(x^a)^b = x^{(a \cdot b)}$. Now these two derivatives must be the same. So

$$px^{p-1} = q \cdot x^{(pq-p)/q} \frac{d}{dx}\{x^{p/q}\}$$

and, rearranging things,

$$\begin{aligned} \frac{d}{dx}\{x^{p/q}\} &= \frac{p}{q} x^{p-1-(pq-p)/q} \\ &= \frac{p}{q} x^{(pq-q-pq+p)/q} \\ &= \frac{p}{q} x^{p/q-1} \end{aligned}$$

So finally

$$\frac{d}{dx}\{x^{p/q}\} = \frac{p}{q} x^{p/q-1} \quad (2.6.2)$$

Notice that this has the same form as Lemma 2.6.9, above, except with $n = p/q$ allowed to be any positive rational number, not just a positive integer.

Example 2.6.14

Example 2.6.15 (Derivative of x^{-m})

In this example we'll use the quotient rule to find the derivative of x^{-m} , for any natural number m .

By the special case of the quotient rule (Corollary 2.4.6) with $g(x) = x^m$ and $g'(x) = mx^{m-1}$

$$\frac{d}{dx}\{x^{-m}\} = \frac{d}{dx}\left\{\frac{1}{x^m}\right\} = -\frac{mx^{m-1}}{(x^m)^2} = -mx^{-m-1}$$

Again, notice that this has the same form as Lemma 2.6.9, above, except with $n = -m$ being a negative integer.

Example 2.6.15

Example 2.6.16

In this example we'll use the quotient rule to find the derivative of $x^{-p/q}$, for any pair of natural numbers p and q . By the special case the quotient rule (Corollary 2.4.6) with $g(x) = x^{p/q}$ and $g'(x) = \frac{p}{q}x^{p/q-1}$,

$$\frac{d}{dx}\{x^{-p/q}\} = \frac{d}{dx}\left\{\frac{1}{x^{p/q}}\right\} = -\frac{\frac{p}{q}x^{p/q-1}}{(x^{p/q})^2} = -\frac{p}{q}x^{-p/q-1}$$

Example 2.6.16

Note that we have found, in Examples 2.2.2, 2.6.14 and 2.6.16, the derivative of x^a for any rational number a , whether 0, positive, negative, integer or fractional. In all cases, the answer is

Corollary 2.6.17 (Derivative of x^a).

Let a be a rational number, then

$$\frac{d}{dx}x^a = ax^{a-1} \quad (2.6.3)$$

We shall show, in Example 2.10.5, that the formula $\frac{d}{dx}x^a = ax^{a-1}$ in fact applies for all real numbers a , not just rational numbers.

Back in Example 2.2.9 we computed the derivative of $\sqrt{x}$ from the definition of the derivative. The above corollary (correctly) gives

$$\frac{d}{dx}x^{1/2} = \frac{1}{2}x^{-1/2}$$

but with far less work.

Here's an (optional) messy example.

Example 2.6.18 (Optional messy example)

Find the derivative of

$$f(x) = \frac{(\sqrt{x} - 1)(2 - x)(1 - x^2)}{\sqrt{x}(3 + 2x)}$$

- As we seen before, the best strategy for dealing with nasty expressions is to break them up into easy pieces. We can think of $f(x)$ as the five-fold product

$$f(x) = f_1(x) \cdot f_2(x) \cdot f_3(x) \cdot \frac{1}{f_4(x)} \cdot \frac{1}{f_5(x)}$$

with

$$f_1(x) = \sqrt{x} - 1 \quad f_2(x) = 2 - x \quad f_3(x) = 1 - x^2 \quad f_4(x) = \sqrt{x} \quad f_5(x) = 3 + 2x$$

- By now, the derivatives of the f_j 's should be easy to find:

$$f'_1(x) = \frac{1}{2\sqrt{x}} \quad f'_2(x) = -1 \quad f'_3(x) = -2x \quad f'_4(x) = \frac{1}{2\sqrt{x}} \quad f'_5(x) = 2$$

- Now, to get the derivative $f'(x)$ we use the n -fold product rule which was developed in Example 2.6.6, together with the special case of the quotient rule (Corollary 2.4.6).

$$\begin{aligned} f'(x) &= f'_1 f_2 f_3 \frac{1}{f_4} \frac{1}{f_5} + f_1 f'_2 f_3 \frac{1}{f_4} \frac{1}{f_5} + f_1 f_2 f'_3 \frac{1}{f_4} \frac{1}{f_5} - f_1 f_2 f_3 \frac{f'_4}{f_4^2} \frac{1}{f_5} - f_1 f_2 f_3 \frac{1}{f_4} \frac{f'_5}{f_5^2} \\ &= \left[\frac{f'_1}{f_1} + \frac{f'_2}{f_2} + \frac{f'_3}{f_3} - \frac{f'_4}{f_4} - \frac{f'_5}{f_5} \right] f_1 f_2 f_3 \frac{1}{f_4} \frac{1}{f_5} \\ &= \left[\frac{1}{2\sqrt{x}(\sqrt{x} - 1)} - \frac{1}{2 - x} - \frac{2x}{1 - x^2} - \frac{1}{2x} - \frac{2}{3 + 2x} \right] \frac{(\sqrt{x} - 1)(2 - x)(1 - x^2)}{\sqrt{x}(3 + 2x)} \end{aligned}$$

The trick that we used in going from the first line to the second line, namely multiplying term number j by $\frac{f'_j(x)}{f_j(x)}$ is often useful in simplifying the derivative of a product of many factors¹⁶.

Example 2.6.18

16 Also take a look at “logarithmic differentiation” in Section 2.10.

►► Stage 2

Q[1]: Suppose $h(t)$ gives the height at time t of the water at a dam, where the units of t are hours and the units of h are meters.

- (a) What is the physical interpretation of the slope of the secant line through the points $(0, h(0))$ and $(24, h(24))$?
- (b) What is the physical interpretation of the slope of the tangent line to the curve $y = h(t)$ at the point $(0, h(0))$?

Q[2]: Suppose $p(t)$ is a function that gives the profit generated by selling t widgets. What is the practical interpretation of $p'(t)$?

Q[3]: $T(d)$ gives the temperature of water at a particular location d metres below the surface. What is the physical interpretation of $T'(d)$? Would you expect the magnitude of $T'(d)$ to be larger when d is near 0, or when d is very large?

Q[4]: $C(w)$ gives the calories in w grams of a particular dish. What does $C'(w)$ describe?

Q[5]: The velocity of a moving object at time t is given by $v(t)$. What is $v'(t)$?

Q[6]: The function $T(j)$ gives the temperature in degrees Celsius of a cup of water after j joules of heat have been added. What is $T'(j)$?

Q[7]: A population of bacteria, left for a fixed amount of time at temperature T , grows to $P(T)$ individuals. Interpret $P'(T)$.

►► Stage 3

Q[8]: You hammer a small nail into a wooden wagon wheel. $R(t)$ gives the number of rotations the nail has undergone t seconds after the wagon started to roll. Give an equation for how quickly the nail is rotating, measured in degrees per second.

Q[9]: A population of bacteria, left for a fixed amount of time at temperature T , grows to $P(T)$ individuals. There is one ideal temperature where the bacteria population grows largest, and the closer the sample is to that temperature, the larger the population is (unless the temperature is so extreme that it causes all the bacteria to die by freezing or boiling). How will $P'(T)$ tell you whether you are colder or hotter than the ideal temperature?

2.4▲ Arithmetic of derivatives - a differentiation toolbox**Exercises**

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

► Stage 1

Q[1]: True or false: $\frac{d}{dx}\{f(x) + g(x)\} = f'(x) + g'(x)$ when f and g are differentiable functions.

Q[2]: True or false: $\frac{d}{dx}\{f(x)g(x)\} = f'(x)g'(x)$ when f and g are differentiable functions.

Q[3]: True or false: $\frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{g^2(x)}$ when f and g are differentiable functions.

Q[4]: Let f be a differentiable function. Use at least three different rules to differentiate $g(x) = 3f(x)$, and verify that they all give the same answer.

► Stage 2

Q[5]: Differentiate $f(x) = 3x^2 + 4x^{1/2}$ for $x > 0$.

Q[6]: Use the product rule to differentiate $f(x) = (2x + 5)(8\sqrt{x} - 9x)$.

Q[7](*) : Find the equation of the tangent line to the graph of $y = x^3$ at $x = \frac{1}{2}$.

Q[8](*) : A particle moves along the x -axis so that its position at time t is given by $x = t^3 - 4t^2 + 1$.

- (a) At $t = 2$, what is the particle's speed?
- (b) At $t = 2$, in what direction is the particle moving?
- (c) At $t = 2$, is the particle's speed increasing or decreasing?

Q[9](*) : Calculate and simplify the derivative of $\frac{2x - 1}{2x + 1}$

Q[10]: What is the slope of the graph $y = \left(\frac{3x + 1}{3x - 2} \right)^2$ when $x = 1$?

Q[11]: Find the equation of the tangent line to the curve $f(x) = \frac{1}{\sqrt{x} + 1}$ at the point $\left(1, \frac{1}{2}\right)$.

► Stage 3

Q[12]: A town is founded in the year 2000. After t years, it has had $b(t)$ births and $d(t)$ deaths. Nobody enters or leaves the town except by birth or death (whoa). Give an expression for the rate the population of the town is growing.

Q[13](*) : Find all points on the curve $y = 3x^2$ where the tangent line passes through $(2, 9)$.

Q[14](*) : Evaluate $\lim_{y \rightarrow 0} \left(\frac{\sqrt{100180 + y} - \sqrt{100180}}{y} \right)$ by interpreting the limit as a derivative.

Q[15]: A rectangle is growing. At time $t = 0$, it is a square with side length 1 metre. Its width increases at a constant rate of 2 metres per second, and its length increases at a constant rate of 5 metres per second. How fast is its area increasing at time $t > 0$?

Q[16]: Let $f(x) = x^2 g(x)$ for some differentiable function $g(x)$. What is $f'(0)$?

Q[17]: Verify that differentiating $f(x) = \frac{g(x)}{h(x)}$ using the quotient rule gives the same answer as differentiating $f(x) = \frac{g(x)}{k(x)} \cdot \frac{k(x)}{h(x)}$ using the product rule and the quotient rule.

2.5▲ Proofs of the arithmetic of derivatives

No exercises for Section [2.5](#)

Jump to [TABLE OF CONTENTS](#).

2.6▲ Using the arithmetic of derivatives - examples

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Spot and correct the error(s) in the following calculation.

$$\begin{aligned}f(x) &= \frac{2x}{x+1} \\f'(x) &= \frac{2(x+1) + 2x}{(x+1)^2} \\&= \frac{2(x+1)}{(x+1)^2} \\&= \frac{2}{x+1}\end{aligned}$$

Q[2]: True or false: $\frac{d}{dx}\{2^x\} = x2^{x-1}$.

►► Stage 2

Q[3]: Differentiate $f(x) = \frac{2}{3}x^6 + 5x^4 + 12x^2 + 9$ and factor the result.

Q[4]: Differentiate $s(t) = 3t^4 + 5t^3 - \frac{1}{t}$.

Q[5]: Differentiate $x(y) = \left(2y + \frac{1}{y}\right) \cdot y^3$.

Q[6]: Differentiate $T(x) = \frac{\sqrt{x} + 1}{x^2 + 3}$.

Q[7](*) : Compute the derivative of $\left(\frac{7x+2}{x^2+3}\right)$.

Q[8]: What is $f'(0)$, when $f(x) = (3x^3 + 4x^2 + x + 1)(2x + 5)$?

Q[9]: Differentiate $f(x) = \frac{3x^3 + 1}{x^2 + 5x}$.

Q[10](*) : Compute the derivative of $\left(\frac{3x^2 + 5}{2 - x}\right)$

Q[11](*) : Compute the derivative of $\left(\frac{2 - x^2}{3x^2 + 5}\right)$.

Q[12](*) : Compute the derivative of $\left(\frac{2x^3 + 1}{x + 2}\right)$.

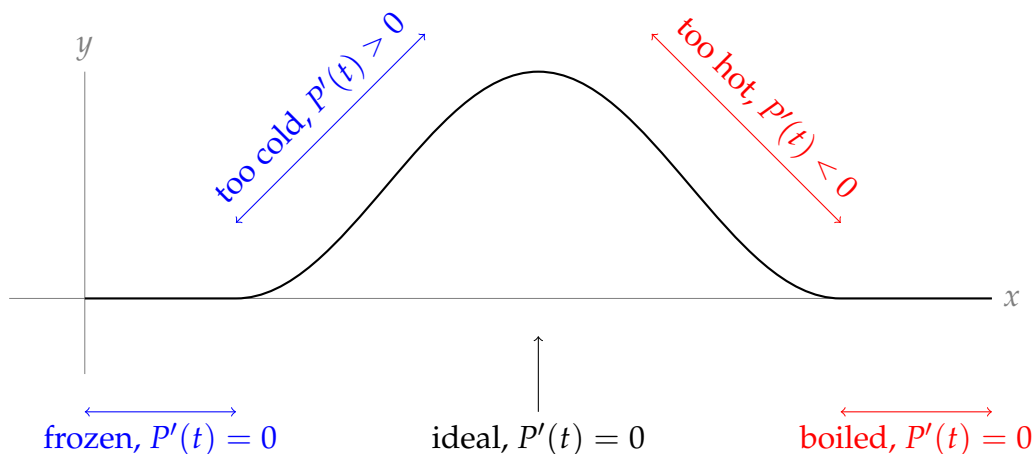
S-7: As usual, it is instructive to think about the definition of the derivative:

$$P'(T) = \lim_{h \rightarrow 0} \frac{P(T+h) - P(T)}{h} \approx \frac{P(T+1) - P(T)}{1} = P(T+1) - P(T).$$

This is the difference in population between two hypothetical populations, raised one degree in temperature apart. So, it is the number of extra individuals that exist in the hotter experiment (with the understanding that this number could be negative, as one would expect in conditions that are hotter than the bacteria prefer). So $P'(T)$ is the number of bacteria added to the colony per degree.

S-8: $R'(t)$ is the rate at which the wheel is rotating measured in rotations per second. To convert to degrees, we multiply by 360: $360R'(t)$.

S-9: If $P'(t)$ is positive, your sample is below the ideal temperature, because adding heat increases the population. If $P'(t)$ is negative, your sample is above the ideal temperature, because adding heat decreases the population. If $P'(t) = 0$, then adding a little bit of heat doesn't change the population, but it's unclear why this is. Perhaps your sample is deeply frozen, and adding heat doesn't change the fact that your population is 0. Perhaps your sample is boiling, and again, changing the heat a little will keep the population constant at "none." But also, at the ideal temperature, you would expect $P'(t) = 0$. This is best seen by noting in the curve below, the tangent line is horizontal at the peak.



Solutions to Exercises 2.4 — Jump to [TABLE OF CONTENTS](#)

S-1: True: this is exactly what the Sum Rule states.

S-2: False, in general. The product rule tells us $\frac{d}{dx}\{f(x)g(x)\} = f'(x)g(x) + f(x)g'(x)$. An easy example of why we can't do it the other way is to take $f(x) = g(x) = x$. Then the equation becomes $\frac{d}{dx}\{x^2\} = (1)(1)$, which is false.

S-3: True: the quotient rule tells us

$$\frac{d}{dx} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)} = \frac{g(x)f'(x)}{g^2(x)} - \frac{f(x)g'(x)}{g^2(x)} = \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{g^2(x)}.$$

S-4: If you're creative, you can find lots of ways to differentiate!

Constant multiple: $g'(x) = 3f'(x)$.

Product rule: $g'(x) = \frac{d}{dx}\{3\}f(x) + 3f'(x) = 0f(x) + 3f'(x) = 3f'(x)$.

Sum rule: $g'(x) = \frac{d}{dx}\{f(x) + f(x) + f(x)\} = f'(x) + f'(x) + f'(x) = 3f'(x)$.

Quotient rule: $g'(x) = \frac{d}{dx} \left\{ \frac{f(x)}{\frac{1}{3}} \right\} = \frac{\frac{1}{3}f'(x) - f(x)(0)}{\frac{1}{9}} = \frac{\frac{1}{3}f'(x)}{\frac{1}{9}} = 9 \left(\frac{1}{3} \right) f'(x) = 3f'(x)$.

All rules give $g'(x) = 3f'(x)$.

S-5: We know, from Examples 2.2.5 and 2.2.9 in the CLP-1 text, that $\frac{d}{dx}x^2 = 2x$ and $\frac{d}{dx}x^{1/2} = \frac{1}{2\sqrt{x}}$. So, by linearity,

$$f'(x) = 3 \cdot 2x + 4 \cdot \frac{1}{2\sqrt{x}} = 6x + \frac{2}{\sqrt{x}}$$

S-6: We have already seen $\frac{d}{dx}\{\sqrt{x}\} = \frac{1}{2\sqrt{x}}$ in Example 2.2.9 of the CLP-1 text. Now:

$$\begin{aligned} f'(x) &= (2)(8\sqrt{x} - 9x) + (2x + 5) \left(\frac{8}{2\sqrt{x}} - 9 \right) \\ &= 16\sqrt{x} - 18x + (2x + 5) \left(\frac{4}{\sqrt{x}} - 9 \right) \\ &= -36x + 24\sqrt{x} + \frac{20}{\sqrt{x}} - 45 \end{aligned}$$

S-7: We already know that $\frac{d}{dx}x = 1$ and $\frac{d}{dx}x^2 = 2x$, so we can compute the derivative of x^3 by writing $x^3 = (x)(x^2)$,

$$\frac{d}{dx}x^3 = \frac{d}{dx}(x)(x^2) = (1)(x^2) + (x)(2x) = 3x^2$$

When this is evaluated at $x = \frac{1}{2}$ we get $\frac{3}{4}$. Since we also compute $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, the equation of the tangent line is

$$y - \frac{1}{8} = \frac{3}{4} \cdot \left(x - \frac{1}{2} \right).$$

S-8: Let $f(t) = t^3 - 4t^2 + 1$. We saw in Question **7** that $\frac{d}{dt}t^3 = 3t^2$. So

$$\begin{aligned}f'(t) &= 3t^2 - 8t & f'(2) &= 3 \times 4 - 8 \times 2 = -4 \\f''(t) &= 6t - 8 & f''(2) &= 6 \times 2 - 8 = 4\end{aligned}$$

and the speed is

$$|f'(t)| = |3t^2 - 8t| \qquad |f'(2)| = |-4| = 4$$

Hence at $t = 2$, **(a)** the particle has speed 4, and **(b)** is moving towards the left. At $t = 2$, $f''(2) > 0$, so f' is increasing. For t near 2, $f'(t)$ is negative so that the speed $|f'(t)| = -f'(t)$ is decreasing. Thus **(c)** the speed is decreasing.

S-9: We can use the quotient rule here.

$$\frac{d}{dx} \left\{ \frac{2x-1}{2x+1} \right\} = \frac{(2x+1)(2) - (2x-1)(2)}{(2x+1)^2} = \frac{4}{(2x+1)^2} = \frac{1}{(x+1/2)^2}$$

S-10: First, we find the y' for general x . Using the corollary to Theorem 2.4.3 and the quotient rule:

$$\begin{aligned}y' &= 2 \left(\frac{3x+1}{3x-2} \right) \cdot \frac{d}{dx} \left\{ \frac{3x+1}{3x-2} \right\} \\&= 2 \left(\frac{3x+1}{3x-2} \right) \left(\frac{(3x-2)(3) - (3x+1)(3)}{(3x-2)^2} \right) \\&= 2 \left(\frac{3x+1}{3x-2} \right) \left(\frac{-9}{(3x-2)^2} \right) \\&= \frac{-18(3x+1)}{(3x-2)^3}\end{aligned}$$

So, plugging in $x = 1$:

$$y'(1) = \frac{-18(3+1)}{(3-2)^3} = -72$$

S-11: We need $f'(1)$, so first we must find $f'(x)$. Since $f(x)$ is the reciprocal of $\sqrt{x} + 1$, we can use the Corollary 2.4.6:

$$f'(x) = \frac{-\frac{d}{dx}\{\sqrt{x} + 1\}}{(\sqrt{x} + 1)^2} = \frac{-\frac{1}{2\sqrt{x}}}{(\sqrt{x} + 1)^2} = \frac{-1}{2\sqrt{x}(\sqrt{x} + 1)^2},$$

$$\text{so } f'(1) = \frac{-1}{2\sqrt{1}(\sqrt{1} + 1)^2} = \frac{-1}{8}.$$

Now, using the point $\left(1, \frac{1}{2}\right)$ and the slope $\frac{-1}{8}$, our tangent line has equation $y - \frac{1}{2} = \frac{-1}{8}(x - 1)$.

S-12: Population growth is rate of change of population. Population in year $2000 + t$ is given by $P(t) = P_0 + b(t) - d(t)$, where P_0 is the initial population of the town. Then $P'(t)$ is the expression we're looking for, and $P'(t) = b'(t) - d'(t)$.

It is interesting to note that the initial population does not obviously show up in this calculation. It would probably affect $b(t)$ and $d(t)$, but if we know these we do not need to know P_0 to answer our question.

S-13: We already know that $\frac{d}{dx}x^2 = 2x$. So the slope of $y = 3x^2$ at $x = a$ is $6a$. The tangent line to $y = 3x^2$ at $x = a, y = 3a^2$ is $y - 3a^2 = 6a(x - a)$. This tangent line passes through $(2, 9)$ if

$$\begin{aligned}9 - 3a^2 &= 6a(2 - a) \\3a^2 - 12a + 9 &= 0 \\a^2 - 4a + 3 &= 0 \\(a - 3)(a - 1) &= 0 \\ \implies a &= 1, 3\end{aligned}$$

The points are $(1, 3)$, $(3, 27)$.

S-14: This limit represents the derivative computed at $x = 100180$ of the function $f(x) = \sqrt{x}$. Since the derivative of $f(x)$ is $\frac{1}{2\sqrt{x}}$, then its value at $x = 100180$ is exactly $\frac{1}{2\sqrt{100180}}$.

S-15: Let $w(t)$ and $l(t)$ be the width and length of the rectangle. Given in the problem is that $w'(t) = 2$ and $l'(t) = 5$. Since both functions have constant slopes, both must be lines. Their slopes are given, and their intercepts are $w(0) = l(0) = 1$. So, $w(t) = 2t + 1$ and $l(t) = 5t + 1$.

The area of the rectangle is $A(t) = w(t) \cdot l(t)$, so using the product rule, the rate at which the area is increasing is $A'(t) = w'(t)l(t) + w(t)l'(t) = 2(5t + 1) + 5(2t + 1) = 20t + 7$ square metres per second.

S-16: Using the product rule, $f'(x) = (2x)g(x) + x^2g'(x)$, so $f'(0) = 0 \cdot g(x) + 0 \cdot g'(x) = 0$. (Since g is differentiable, g' exists.)

S-17:

First expression, $f(x) = \frac{g(x)}{h(x)}$:

$$f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{h^2(x)}$$

Second expression, $f(x) = \frac{g(x)}{k(x)} \cdot \frac{k(x)}{h(x)}$:

$$\begin{aligned}
 f'(x) &= \left(\frac{k(x)g'(x) - g(x)k'(x)}{k^2(x)} \right) \left(\frac{k(x)}{h(x)} \right) + \left(\frac{g(x)}{k(x)} \right) \left(\frac{h(x)k'(x) - k(x)h'(x)}{h^2(x)} \right) \\
 &= \frac{k(x)g'(x) - g(x)k'(x)}{k(x)h(x)} + \frac{g(x)h(x)k'(x) - g(x)k(x)h'(x)}{k(x)h^2(x)} \\
 &= \frac{h(x)k(x)g'(x) - h(x)g(x)k'(x)}{k(x)h^2(x)} + \frac{g(x)h(x)k'(x) - g(x)k(x)h'(x)}{k(x)h^2(x)} \\
 &= \frac{h(x)k(x)g'(x) - h(x)g(x)k'(x) + g(x)h(x)k'(x) - g(x)k(x)h'(x)}{k(x)h^2(x)} \\
 &= \frac{h(x)k(x)g'(x) - g(x)k(x)h'(x)}{k(x)h^2(x)} \\
 &= \frac{h(x)g'(x) - g(x)h'(x)}{h^2(x)}
 \end{aligned}$$

and this is exactly what we got from differentiating the first expression.

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Solutions to Exercises 2.6 — Jump to [TABLE OF CONTENTS](#)

S-1: In the quotient rule, there is a minus, not a plus. Also, $2(x+1) + 2x$ is not the same as $2(x+1)$.

The correct version is:

$$\begin{aligned}
 f(x) &= \frac{2x}{x+1} \\
 f'(x) &= \frac{2(x+1) - 2x}{(x+1)^2} \\
 &= \frac{2}{(x+1)^2}
 \end{aligned}$$

S-2: False: Lemma 2.6.9 tells us that, for a constant n , $\frac{d}{dx}\{x^n\} = nx^{n-1}$. Note that the base x is the variable and the exponent n is a constant. In the equation given in the question, the base 2 is a constant, and the exponent x is the variable: this is the opposite of the situation required by Lemma 2.6.9. So Lemma 2.6.9 *does not apply*.

We do not yet know how to differentiate 2^x . We'll learn about it in Section 2.7. There we will see that the derivative is $(\log_e 2) 2^x$ *not* $x 2^{x-1}$.

S-3: $f(x) = \frac{2}{3}x^6 + 5x^4 + 12x^2 + 9$ is a polynomial:

$$\begin{aligned}f'(x) &= 4x^5 + 20x^3 + 24x \\&= 4x(x^4 + 5x^2 + 6) \\&= 4x((x^2)^2 + 5(x^2) + 6) \\&= 4x(x^2 + 2)(x^2 + 3)\end{aligned}$$

S-4: We can rewrite slightly to make every term into a power of t :

$$\begin{aligned}s(t) &= 3t^4 + 5t^3 - t^{-1} \\s'(t) &= 4 \cdot 3t^3 + 3 \cdot 5t^2 - (-1) \cdot t^{-2} \\&= 12t^3 + 15t^2 + \frac{1}{t^2}\end{aligned}$$

S-5: We could use the product rule here, but it's easier to simplify first. Don't be confused by the role reversal of x and y : x is the name of the function, and y is the variable.

$$\begin{aligned}x(y) &= \left(2y + \frac{1}{y}\right) \cdot y^3 \\&= 2y^4 + y^2 \\x'(y) &= 8y^3 + 2y\end{aligned}$$

S-6: We've already seen that $\frac{d}{dx}\{\sqrt{x}\} = \frac{1}{2\sqrt{x}}$, but if you forget this formula it is easy to figure out: $\sqrt{x} = x^{1/2}$, so $\frac{d}{dx}\{\sqrt{x}\} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$.

Using the quotient rule:

$$\begin{aligned}T(x) &= \frac{\sqrt{x} + 1}{x^2 + 3} \\T'(x) &= \frac{(x^2 + 3) \left(\frac{1}{2\sqrt{x}}\right) - (\sqrt{x} + 1)(2x)}{(x^2 + 3)^2}\end{aligned}$$

S-7: We use quotient rule:

$$\frac{(x^2 + 3) \cdot 7 - 2x \cdot (7x + 2)}{(x^2 + 3)^2} = \frac{21 - 4x - 7x^2}{(x^2 + 3)^2}$$

S-8: Instead of multiplying to get our usual form of this polynomial, we can use the product rule. If $f_1(x) = 3x^3 + 4x^2 + x + 1$ and $f_2(x) = 2x + 5$, then $f_1'(x) = 9x^2 + 8x + 1$ and $f_2'(x) = 2$. Then

$$\begin{aligned} f'(0) &= f_1'(0)f_2(0) + f_1(0)f_2'(0) \\ &= (1)(5) + (1)(2) = 7 \end{aligned}$$

S-9: Using the quotient rule,

$$f'(x) = \frac{(x^2 + 5x)(9x^2) - (3x^3 + 1)(2x + 5)}{(x^2 + 5x)^2} = \frac{3x^4 + 30x^3 - 2x - 5}{(x^2 + 5x)^2}$$

S-10: We use quotient rule:

$$\frac{(2 - x)(6x) - (3x^2 + 5)(-1)}{(2 - x)^2} = \frac{-3x^2 + 12x + 5}{(x - 2)^2}$$

S-11: We use quotient rule:

$$\frac{(3x^2 + 5)(-2x) - (2 - x^2)(6x)}{(3x^2 + 5)^2} = \frac{-22x}{(3x^2 + 5)^2}$$

S-12: We use quotient rule:

$$\frac{6x^2 \cdot (x + 2) - (2x^3 + 1) \cdot 1}{(x + 2)^2} = \frac{4x^3 + 12x^2 - 1}{(x + 2)^2}$$

S-13: The derivative of the function is

$$\frac{(1 - x^2) \cdot \frac{1}{2\sqrt{x}} - \sqrt{x} \cdot (-2x)}{(1 - x^2)^2} = \frac{(1 - x^2) - 2x \cdot (-2x)}{2\sqrt{x}(1 - x^2)^2}$$

The derivative is undefined if either $x < 0$ or $x = 0, \pm 1$ (since the square-root is undefined for $x < 0$ and the denominator is zero when $x = 0, 1, -1$). Putting this together — the derivative exists for $x > 0, x \neq 1$.

S-14: Using the product rule seems faster than expanding.

$$\begin{aligned}f'(x) &= \frac{d}{dx} \{3\sqrt[5]{x} + 15\sqrt[3]{x} + 8\} (3x^2 + 8x - 5) + (3\sqrt[5]{x} + 15\sqrt[3]{x} + 8) \frac{d}{dx} \{3x^2 + 8x - 5\} \\&= \frac{d}{dx} \left\{3x^{\frac{1}{5}} + 15x^{\frac{1}{3}} + 8\right\} (3x^2 + 8x - 5) + (3\sqrt[5]{x} + 15\sqrt[3]{x} + 8) \frac{d}{dx} \{3x^2 + 8x - 5\} \\&= \left(\frac{3}{5}x^{-\frac{4}{5}} + 5x^{-\frac{2}{3}}\right) (3x^2 + 8x - 5) + (3\sqrt[5]{x} + 15\sqrt[3]{x} + 8) (6x + 8)\end{aligned}$$

S-15: To avoid the quotient rule, we can divide through the denominator:

$$\begin{aligned}f(x) &= \frac{(x^2 + 5x + 1)(\sqrt{x} + \sqrt[3]{x})}{x} = (x^2 + 5x + 1) \frac{(\sqrt{x} + \sqrt[3]{x})}{x} \\&= (x^2 + 5x + 1)(x^{-1/2} + x^{-2/3})\end{aligned}$$

Now, product rule:

$$f'(x) = (2x + 5)(x^{-1/2} + x^{-2/3}) + (x^2 + 5x + 1) \left(-\frac{1}{2}x^{-3/2} - \frac{2}{3}x^{-5/3} \right)$$

(If you simplified differently, or used the quotient rule, you probably came up with a different-looking answer. There is only one derivative, though, so all correct answers will look the same after sufficient algebraic manipulation.)

S-16: The question asks us to find where $f'(x) = 0$ and $f(x)$ exists. We can use the formula for the derivative of a reciprocal, Corollary 2.4.6:

$$\begin{aligned}f'(x) &= \frac{-\frac{d}{dx} \left\{ \frac{1}{5}x^5 + x^4 - \frac{5}{3}x^3 \right\}}{\left(\frac{1}{5}x^5 + x^4 - \frac{5}{3}x^3 \right)^2} \\&= \frac{-(x^4 + 4x^3 - 5x^2)}{\left(\frac{1}{5}x^5 + x^4 - \frac{5}{3}x^3 \right)^2} \\&= \frac{-x^2(x^2 + 4x - 5)}{\left(\frac{1}{5}x^5 + x^4 - \frac{5}{3}x^3 \right)^2} \\&= \frac{-x^2(x + 5)(x - 1)}{\left(\frac{1}{5}x^5 + x^4 - \frac{5}{3}x^3 \right)^2}\end{aligned}$$

So our candidates for x -values where $f'(x) = 0$ are $x = 0$, $x = -5$, and $x = 1$. However, we need to check that f exists at these places: $f(0)$ is undefined (and $f'(0)$ doesn't exist). So $f'(x) = 0$ only when $x = -5$ and $x = 1$.

S-17: Denote by m the slope of the common tangent, by (x_1, y_1) the point of tangency with $y = x^2$, and by (x_2, y_2) the point of tangency with $y = x^2 - 2x + 2$. Then we must have

$$y_1 = x_1^2 \quad y_2 = x_2^2 - 2x_2 + 2 \quad m = 2x_1 = 2x_2 - 2 = \frac{y_2 - y_1}{x_2 - x_1}$$

From the “ m ” equations we get $x_1 = \frac{m}{2}$, $x_2 = \frac{m}{2} + 1$ and

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= y_2 - y_1 \\ &= x_2^2 - 2x_2 + 2 - x_1^2 \\ &= (x_2 - x_1)(x_2 + x_1) - 2(x_2 - 1) \\ &= \left(\frac{m}{2} + 1 - \frac{m}{2}\right) \left(\frac{m}{2} + 1 + \frac{m}{2}\right) - 2\left(\frac{m}{2} + 1 - 1\right) \\ &= (1)(m + 1) - 2\frac{m}{2} \\ &= 1 \end{aligned}$$

$$\text{So, } m = 1, \quad x_1 = \frac{1}{2}, \quad y_1 = \frac{1}{4}, \quad x_2 = \frac{3}{2}, \quad y_2 = \frac{9}{4} - 3 + 2 = \frac{5}{4}$$

An equation of the common tangent is $y = x - \frac{1}{4}$.

S-18: The line $y = mx + b$ is tangent to $y = x^2$ at $x = \alpha$ if

$$2\alpha = m \text{ and } \alpha^2 = m\alpha + b \iff m = 2\alpha \text{ and } b = -\alpha^2$$

The same line $y = mx + b$ is tangent to $y = -x^2 + 2x - 5$ at $x = \beta$ if

$$\begin{aligned} -2\beta + 2 &= m \text{ and } -\beta^2 + 2\beta - 5 = m\beta + b \\ \iff m &= 2 - 2\beta \text{ and } b = -\beta^2 + 2\beta - 5 - (2 - 2\beta)\beta = \beta^2 - 5 \end{aligned}$$

For the line to be simultaneously tangent to the two parabolas we need

$$m = 2\alpha = 2 - 2\beta \text{ and } b = -\alpha^2 = \beta^2 - 5$$

Substituting $\alpha = 1 - \beta$ into $-\alpha^2 = \beta^2 - 5$ gives $-(1 - \beta)^2 = \beta^2 - 5$ or $2\beta^2 - 2\beta - 4 = 0$ or $\beta = -1, 2$. The corresponding values of the other parameters are $\alpha = 2, -1$, $m = 4, -2$ and $b = -4, -1$. The two lines are $y = 4x - 4$ and $y = -2x - 1$.

